

Math AA HL at KCA - Chapter 14 Notes

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1 Differentiation by First Principles

The derivative of a function is defined using limits as

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

2 Differentiation Rules

- $\frac{d}{dx} x^n = nx^{n-1}$
- $\frac{d}{dx} \sin x = \cos x$
- $\frac{d}{dx} \cos x = -\sin x$
- $\frac{d}{dx} \tan x = \frac{1}{\cos^2 x} = \sec^2 x = 1 + \tan^2 x$
- $\frac{d}{dx} \csc x = -\csc x \cot x$
- $\frac{d}{dx} \sec x = \tan x \sec x$
- $\frac{d}{dx} \cot x = -\csc^2 x$
- $\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx} \arccos x = -\frac{1}{\sqrt{1-x^2}} = -(\arcsin x)'$
- $\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$
- $\frac{d}{dx} e^x = e^x$
- $\frac{d}{dx} a^x = a^x \cdot \ln a$
- Product rule: $\frac{d}{dx} f(x)g(x) = f'(x)g(x) + f(x)g'(x)$
- Quotient rule: $\frac{d}{dx} \frac{f(x)}{g(x)} = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$
- Chain rule: $\frac{d}{dx} f(g(x)) = f'(g(x)) \times g'(x)$

Basic differentiation rules are usually combined through the chain rule.

3 Equations of Tangents and Normals

Given a function $f(x)$, the equation of the tangent line at $x = a$ is

$$y = f'(a)(x - a) + f(a)$$

The normal at $x = a$ is given by

$$y = -\frac{1}{f'(a)}(x - a) + f(a)$$

4 Increasing and Decreasing Functions over an Interval

- $f'(a) = 0 \iff (a, f(a))$ is a *stationary point*
- $f'(a) > 0$ for $x \in (a, b)$, then $f(x)$ is *increasing* in the interval (a, b)
- $f'(a) < 0$ for $x \in (a, b)$, then $f(x)$ is *decreasing* in the interval (a, b)

5 Second Order Derivatives

The second-order derivative of $y = f(x)$ can be written as

$$f''(x) = \frac{d^2y}{dx^2}$$

Interpreting the second-order derivative

- $f''(a) = 0 \iff (a, f(a))$ is a point of inflection, if the graph alters its concavity (concave to convex or vice versa) at that point
- $f''(a) > 0 \iff$ convex/concave up at $x = a$; for a turning point, it is then a *minimum point*
- $f''(a) < 0 \iff$ concave down at $x = a$; for a turning point, it is then a *maximum point*

6 Sketching Gradient Functions

On $f(x)$	On $f'(x)$
Turning point	Cuts the x -axis
Inflection point	Turning point
Positive gradient	Above x -axis
Negative gradient	Below x -axis
Vertical asymptote	Vertical asymptote
Horizontal asymptote	Horizontal asymptote $y = 0$

7 Implicit Differentiation

Treat all other variables as a function of the variable with respect to which the entire expression is differentiated. In the end, isolate $\frac{dy}{dx}$

E.g. differentiate the following

$$x^2 + y^2 = 4$$

$$2x + 2y \frac{dy}{dx} = \frac{d}{dx} 4$$

$$x + y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{y}$$